

We will see –

1. How to apply Baye's rule in practice?
2. How to atleast approximately find class conditionals $p(x|\omega_i)$?
3. Finally, this leads to a very simple classifier called k nearest neighbor classifier.

How to find class conditionals $p(X|\omega_i)$?

Let $\Omega = \{\omega_1, \dots, \omega_c\}$ be the set of classes.

Let $\mathcal{D} = \{(X_1, y_1), \dots, (X_n, y_n)\}$ be the training set.

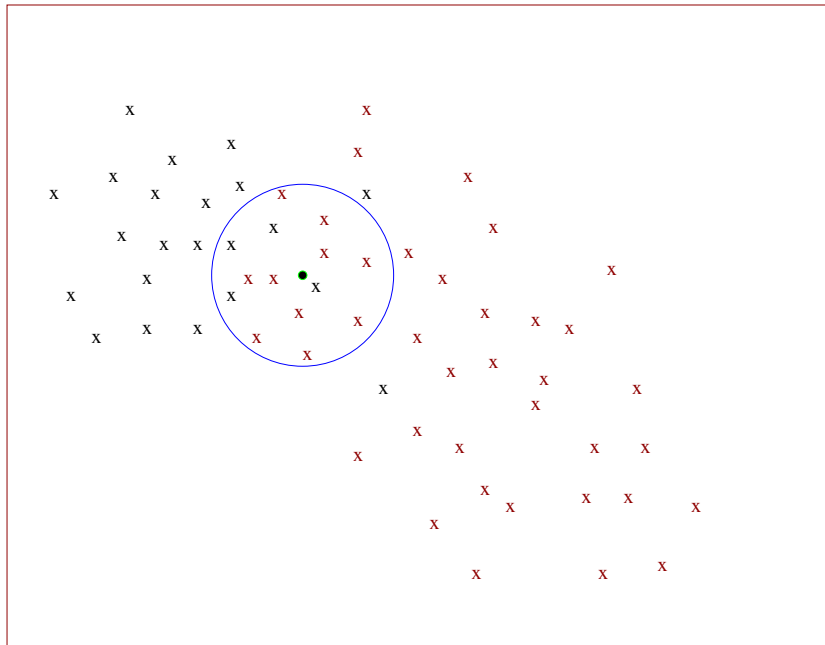
Here $y_i \in \Omega$ is the class label for pattern X_i .

To find $p(X|\omega_i)$, do the following:

1. Draw a small hyper-sphere at X . Let its volume be V .
2. Count the number of training patterns which belongs to class ω_i that are falling in the sphere. Let this be k_i .
3. Let n_i be the total number of training patterns that belongs to class ω_i .
4. Then an estimate of $p(X|\omega_i)$ which is $\hat{p}(X|\omega_i)$ is:

$$\hat{p}(X|\omega_i) = \frac{k_i/n_i}{V}$$

Consider a two class problem. Classes are *black* and *red*



Let X be the centre of the sphere whose volume is 2 units.

$$\hat{p}(X \mid \text{class is } \textit{black}) = 4/(20 * 2) = 1/10.$$

$$\hat{p}(X \mid \text{class is } \textit{red}) = 10/(40 * 2) = 1/8.$$

How to find $\hat{p}(X)$? In this example what is its value?

An estimate of priors would be:

$$\hat{P}(\omega_i) = \frac{n_i}{n}$$

Posterior

$$\hat{P}(\omega_i|X) = \frac{\frac{k_i}{n_i} \times \frac{n_i}{n}}{p(X)} = \frac{k_i/n}{p(X)}$$

Predicted class is according to

$$\begin{aligned} & \text{max. in } \{\hat{P}(\omega_1|X), \dots, \hat{P}(\omega_c|X)\} \\ &= \text{max. in } \left\{ \frac{k_1/n}{p(X)}, \dots, \frac{k_c/n}{p(X)} \right\} \\ &= \text{max. in } \{k_1, \dots, k_c\} \end{aligned}$$

That is, count the number of patterns for each class that are falling in the sphere, and decide the class for which the count is maximum.

Still one problem is: how to decide the radius for the sphere?

Instead of this, for a given value k , we can draw a sphere which is just big enough to hold k nearest neighbors of X .

This classifier is called the *k -nearest neighbor classifier* (*k -NNC*).

For a given test pattern X which is to be classified, do:

1. Find k nearest neighbors of X .
2. Among these k nearest neighbors, according to majority voting decide the predicted class label.

k-NNC is an approximation of Bayes classifier!

Mathematically it can be shown that when $n \rightarrow \infty$, $k \rightarrow \infty$ and $k/n \rightarrow 0$, k-NNC is exactly the Bayes classifier.

When $k = 1$, k-NNC is simply called the *nearest neighbor classifier (NNC)*.

In our example with *black* and *red* classes with $k = 14$ what is the prediction?

Similarly for NNC what is the prediction?

Still there is one problem with k -NNC

That is, how to decide the value of k ?

For this, there is a practical method called *cross validation*.

One more issue is : *How to find the distances?*

There are many distance functions like Euclidean distance.